

IKIP: Comprehensive System Testing Queries

Use the following 22 queries to comprehensively test the Intelligence Chat, Knowledge Graph, and Compliance modules. These queries are specifically designed around the 15 provided test documents.

Category 1: Fact Retrieval (RAG & Semantic Search)

- 1. What is the minimum thickness found on Pressure Vessel V-205 during its annual inspection, and what is the retirement thickness?
- 2. Operators reported a drop in cooling efficiency on Heat Exchanger E-301 back in August. What was the root cause discovered via thermal imaging?
- 3. What PPE is explicitly mandatory when handling Class 3 flammable liquids or Class 8 corrosives according to the Safety Manual?
- 4. What were the exact findings and immediate actions taken in Incident Report IR-2025-017 regarding the P101 seal failure?
- 5. According to the SOP for Pump P-101, what are the LOTO (Lockout/Tagout) and isolation steps required before beginning a seal replacement?
- 6. What is the design pressure and operating temperature specified in the technical specifications of the E-301 Heat Exchanger?
- 7. In the annual equipment inspection report, were there any critical defects noted across the plant that require immediate shutdown?
- 8. List all the maintenance actions that were performed on the compressor station in Q3 of 2025 according to the work order logs.

Category 2: Cross-Document Synthesis (Graph + RAG)

- 1. Summarize the entire maintenance and incident history of Pump P-101 by cross-referencing its incident report and the annual maintenance logs.
- 2. How does the root cause of the E-301 Heat Exchanger's cooling drop relate to its original technical specifications?
- 3. Are there any discrepancies between the safety procedures followed during the P-101 seal failure incident and the official SOP for Pump P-101?
- 4. Which personnel are listed as authors, inspectors, or approvers across the 2025 inspection reports and incident logs?

- 5. Which documents reference Pump P-101 and what role does the pump play in them?

Category 3: Vision & Drawing Analysis

- 1. Analyze the P&ID; for the compressor station drawing. What are the main equipment tags present?
- 2. In the hydraulic schematic drawing, what types of control valves and actuators are shown?
- 3. Does the HVAC layout diagram show any return air ducts connected to the main Air Handling Unit?
- 4. Identify all pump tags present in the 'pid_pump_system' drawing and describe how they are connected.
- 5. What is the drawing number, revision, and unit area of the mechanical isometric piping drawing?